DESIGNING INTERVENTIONS AS LEARNING SPACES

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ABSTRACT

When designing for sustainability, design as a practice needs an altered position. Designing learning spaces with a practice-oriented design can be such an approach. In three mobility interventions that were design-led practice interventions, this design practice was tried out. Furthermore, in these interventions, conditions for sustainable mobility practices were studied. As a suggested design for sustainability approach, aiming to transform socio-technical systems, a design focus on enabling learning spaces where collaborative sensemaking takes place is proposed. However, designing such learning spaces requires careful crafting and designing with altered design postures.

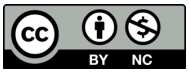
INTRODUCTION

Many approaches are needed to tackle sustainability issues, including behavioural changes, technological innovations, market-driven solutions, and policy-level regulations. There is also a need to understand how design can support such processes and what is required of design activities to assist in sustainability transitions. This publication addresses the research question: *What re-formulations of design as a practice are needed when designing conditions for sustainable mobility practices and supporting transformative learning?* In three mobility interventions, design tools, methods and approaches were used to explore how conditions for sustainable mobility practices could be designed.

If more sustainable living is to become possible, substantial changes will be necessary. However, people's possibilities to live more sustainably are not straightforward since individual behaviours are embedded in institutional contexts (Jackson, 2005). A sustainable life is, therefore, not only about individual choices and consumption behaviours but also about the connecting systems and structures. A sustainable lifestyle is a cluster of habits embedded in society facilitated by norms, institutions, and infrastructures. Taking social practices as a unit of analysis can enable engaging in change and transformations of these lifestyles and structures (Spaargaren, 2003). As Spaargaren (ibid.) argues, social practices, such as travel practices, are constructed in the middle of lifestyles and systems of provision. Social practices as the unit of analysis mean acknowledging that individual responsibilities and possibilities directly relate to how well the systems of provision can support and link with the everyday of the individual. Thereby, systems can enable but also obstruct people from living more sustainably. When using practices as a design unit and with a sustainability agenda, design efforts can be made to provide more sustainable systems, and on the lifestyle side, by encouraging more sustainability-related values.

BACKGROUND

In this publication, the design processes in three mobility interventions are presented. Design practice and research were included as part of these research projects to study conditions for sustainability practices and investigate how these conditions could be designed. However, alternative design approaches are needed to be able to do so (Hesselgren, 2019). Design for sustainability includes several approaches ranging from those that focus on innovation at the product level to those that design for system-level innovation (Ceschin & Gaziulusoy, 2016). Different design approaches for sustainability have various purposes. For example, design for sustainable behaviour aims for behavioural change through interventions which include design



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innovations to support the adoption of sustainable innovations. Thereby, supporting adoptions of innovation comes to the forefront in this approach. On the other hand, sustainable product-service system design aims to transform the production-consumption systems through business model innovation, whereby the business model comes to the forefront. When instead engaging in design for social innovation, the aim is to assist with social change without viewing technological change as a pre-determinant of change. In this case, the focus of design becomes to be an assistant. Additionally, when designing for sustainability transitions, the aim is for the transformation of socio-technical systems through innovations at the technological, social, organisational, and institutional levels. Thereby, design becomes much broader as the socio-technical systems are the centre of attention. Socio-technical systems and understanding how these are configured, and potentially can be reconfigured, are relevant entry points for design (Joore & Brezet, 2015). When engaging in socio-technical systems as configurations, and wanting to design for sustainability, designers can imagine and materialise different and less resource-intensive ways of good living.

In practice-oriented design, the design focuses on the practices (Kuijer, 2014; Kujier & Jong, 2012). Here, to start with, people's everyday practices are the unit of analysis, whereby it becomes a matter of investigating how these have evolved and how they are performed (Kuijer & Jong, 2011). This analysis aims to find opportunities for changing practices by identifying possible interventions. Such a design mindset is proactive rather than depictive, which can be considered the more traditional view in social practice theories (ibid.). A proactive approach, which is a designerly way of engaging in practices, can be to run experiments, living labs or interventions (Kuijer & Jong, 2011; Strengers & Maller, 2015). With such a design approach, attention is given to how materials, meanings and skills are configured together and how links between these elements are made. When unfamiliar elements are introduced, such as new materials, a "crisis of routines" (Reckwitz, 2002) can be caused, and existing practices can be reconfigured into new versions. Such reconfigurations can be "innovations-in-waiting" or "proto-practices" (Shove and Pantzar, 2005). Designing interventions that cause reconfigurations of sociotechnical systems, specifically for proto-practices to emerge, can also be helpful as a way to understand how to make practices more sustainable (Hesselgren, 2019). The reconfigurations are not limited to including material elements but can also be introduced to enable new meanings and skills to develop. Thereby, they can become opportunities for people to renegotiate links within their practices and possibly re-orient their practices into more sustainable ones. Such practice interventions are helpful for taking snapshots of practices and reflecting on how these are performed.

In general, interventions with sustainability agendas can be carried out differently. Strengers and Maller (2015) summarise dominant views on these. First, interventions at the individual level aim to modify individual behaviour, such as nudging people to make specific choices. Second, there are interventions through markets where incentives and prices are used through rational economic models to persuade people in particular directions. Third, there are interventions that, through the launch of new technologies, possibly disrupt people and pull them in a specific direction. All of these can be carried out for sustainability purposes, and designers can engage in all three types of interventions. However, when interventions have a practice focus, different questions can be asked, and alternative approaches can be used to realise what would be needed for transformations. Design can inspire innovative approaches for such interventions, whereby the interventions in themselves can be used as learning spaces (Browne et al., 2015; Maller & Strengers, 2015; Pink and Leder Mackey, 2015). Spurling and McMeekin (2015) discuss how the three approaches: recrafting, or substituting practices, or changing how practices interlock, can be used from a policy perspective for practice interventions to support sustainable mobility practices to emerge.

Hesselgren (2019) suggests using practices as design material and designing learning spaces with opportunities for understanding and re-envisioning practices. When designing learning spaces, these can be considered arenas of experimentation that support learning by doing and reflections in actions. Furthermore, such learning spaces invite single-, double- and third-loop learning, where participants are encouraged to be challenged and to challenge what might be taken for granted. Such nodes of reflections can trigger new learning loops. Cyclical learning with iterative testing and reflecting are needed and valuable when addressing complex sustainability issues.

DESIGNING MOBILITY INTERVENTIONS

In this section, three cases of mobility interventions are described. These were design-led practice interventions for sustainability carried out through the proposed design engagements RE-DO Design (Hesselgren, 2019). The interventions were made in everyday life through purposefully designed systems that reconfigured practices with the intent to explore these for sustainability purposes. To do so, all interventions had a focus on proto-practices. First, the interventions were set up by reconfiguring the participants' everyday lives during the research projects. Second, there was attention

to allowing these proto-practices to be performed. Third, the incubated proto-practices were explored. Fourth, suggestions of how the proto-practices could be refined to become more mainstream versions of what a few had managed to embrace or tried out. The four intervention stages of RE-DO Design are presented in Figure 1.

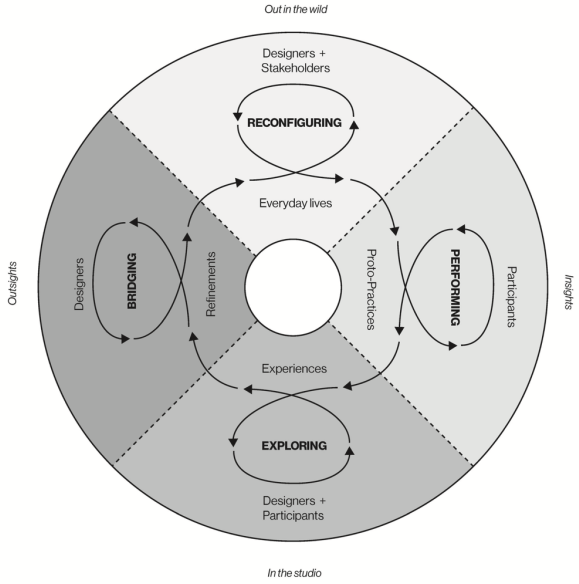


Figure 1: RE-DO Design Doings with four iterative design stages and learning loops used in the interventions – Reconfiguring everyday lives, Performing proto-practices, Exploring experiences, and Bridging refinements (from Hesselgren, 2019).

With RE-DO Design Doings, there is attention to forming learning spaces through design engagements in the four different loops of the interventions. After reconfiguring everyday lives, the intervention participants build first-order learning by performing proto-practices and developing insights. In this stage, one of the design engagements is to capture these insights. Next, second-order learning and interpretations develop as the participants' experiences are explored. Glitches and failures are pinpointed and used as a design material and potential for reoriented proposals for the bridging. With reflections on the learning, third-order learning can develop. Here, deciding on the representations of these potentials is one of the design engagements.

CASE 1

The first case is a research project in Stockholm where three families were engaged to replace their cars with light-electric vehicles: electric bikes, electric box bikes, electric scooters, and an electric four-wheeled motorcycle, for one year, and they also used active mobility and public transport instead of using their cars (see Figure 2). For more details regarding the research project A Car-free Year, see Hasselqvist and Hesselgren (2019).



Figure 2: Examples of the light-electric vehicles used for reconfiguring families' everyday lives.

The participants who changed to car-free living engaged in proto-practices when they enrolled in new travel practices with light-electric vehicles. The families were supported with the material element (the vehicles) and knowledge, such as how to dress for winter cycling. This enabled them to develop the necessary skills for the winter cycling practice. During the trial year, the participants also had free access to repair services to fix flat tyres and mend their equipment when necessary. This support proved essential for them to remain in the winter biking practice.

Furthermore, the proto-practices developed during the intervention were often bundled with several related practices. For example, doing weekly grocery shopping was bundled with travel practices. Replacing using a car to get to the grocery shopping by instead using light-electric vehicles also affected the shopping practice. Car-dependant practices related to commuting, regular weekday trips and irregular vacation trips were reflected upon in the interviews with the participants.

The families' travel practices were explored through monthly interviews in the participant's homes. To encourage reflection and to obtain some of the tacit knowledge the participants developed, the adults in the families kept a travel diary where they registered their experiences, including emotional aspects and thoughts on their new travel practices. Moreover, they wrote on a scorecard the car trips they were obliged to do when they could not find any other solution (see Figure 3).

Twelve trips were allowed during the year. This simple learning device proved helpful in learning about when it was not possible to be car-free. The travel diary sparked discussions by focusing on learning from the participants' experiences. It was used as a trigger material in the interviews to capture the participants' experiences while enrolling in the proto-practices. As the months went by, it became easier for the participants to discuss the values of living car-free from many different perspectives. For example, they raised that they had found it difficult to challenge existing norms on the meanings of good parenthood, particularly with other families with children. This became clear, for example, when the participant's children used public transport more regularly on their own, even from a young age. The parents felt this challenged the norm of good parenthood in their communities.



Figure 3: Example of travel diary and car trip scorecard used for the participants to capture their experiences, emotions and insights while performing the proto-practices.

In the following stage, concepts for supporting car-free living were designed based on the families' positive and negative experiences. The concepts were developed collaboratively, where the families were engaged in co-design workshops (see Figure 4). Since several of the concepts pointed to that collaborations between several stakeholders were essential, there were also co-design workshops carried out with stakeholders from civil society and the public and private sectors.



Figure 4: Example from a workshop with family members co-creating meanings of potential concepts supporting car-free living.

The concepts were visualised as two-dimensional images that aimed to balance the concrete aspects of the families' experiences with more abstract ideas to stimulate thinking further and trigger new ideas of future potential (see Figure 5). To complement the two-dimensional visuals, two videos were made with the voices of the families and the research team sharing stories and reflecting on what car-free living was like.

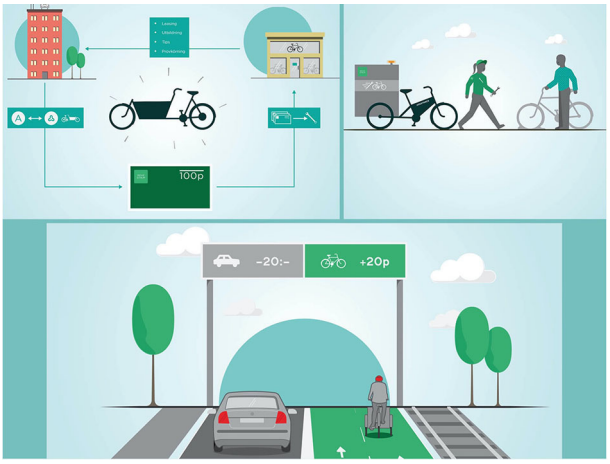


Figure 5: Examples of supporting concepts for car-free living.

The visuals and films were designed to communicate and bridge the knowledge from the study in several meetings at local, regional, and national levels to discuss future sustainable mobility solutions. These discussions were, in some cases, very detailed, for example, concerning the colours of national road signs, but in other cases, more focused on significant implementation challenges of planning cities for sustainable urban mobility.

CASE 2

The second case is a collaborative research project that engaged partners from academia, the private and public sectors, and the third sector. A pool of ten light-electric vehicles, a four-wheeled motorcycle (see Figure 6), was set up as a mobility service system at two workplaces and used there for 6 to 9 months. For more details regarding the research project KTH Mobility Pool, see Sopjani et al. (2020).



Figure 6: KTH Mobility Pool with its vehicles at one of the workplaces.

These interventions engaged two types of users: caretakers of the vehicles and daytime users in the two workplaces. The caretakers took on the responsibility of replacing their commuting with a fossil-fuelled car to work with this vehicle instead. Moreover, daytime users could use this vehicle and the service system during office hours to attend meetings or conduct errands at lunchtime. The project explored the intervention in the routines of going to work and travelling short distances at and around work. Both types of users enrolled in proto-practices as they replaced using their private cars for these travel practices.

The intervention into the caretaker's everyday life was of primary interest since this allowed exploration of both commuting practices and how they engaged in other travel practices outside work. Service design methods were used to engage the caretakers in the discussion, such as mapping their journeys, associated values and emotions, and experiences from adding a new service to their everyday life (see Figure 7).



Figure 7: Example from co-design workshop mapping the caretakers' journeys through the service system.

Individual learning was supported by sending the participants sensitising questions before meeting them for interviews or workshops. With such questions, the participants reflected on their previous experiences, both car-related and newly acquired mobility practices (the proto-practices). The reflections also revealed their tacit knowledge of being a caretaker, engaging in the system as a shared service user, and driving an electric vehicle. Furthermore, learning took place between them as colleagues, and with some daytime users, they enrolled in communities of a new mobility practice. Some expressed how they became ambassadors of sustainable mobility. For some, this fitted very well with their personality and corporate roles. The distinctive-looking vehicle was considered in line with themselves as expressive, and they enjoyed the attention that the vehicle brought them.

Moreover, when this also was in line with their work roles as, for example, sustainability managers, these different perspectives of values enforced each other. However, there were also cases when the attention-seeking vehicle did not align with the caretaker's values. In those cases, it was instead the functional values and practical benefits that these participants promoted.

The insights from the caretaker's experiences of living with the shared mobility service system as their primary source for commuting trips were later brought to the stakeholders involved in the research project. The caretaker's experiences were brought into co-design workshops, using different design methods such as persona descriptions and films that the users had made to present their experiences. This brought their voices

into the decision-making and supported further developments of the mobility service system. In generative sessions, scenarios were co-created by the stakeholders engaged in these workshops (see Figure 8). Discussions of how it could be possible to support sustainable mobility practices further were presented at each workplace. In this bridging process, a particular focus on the learning spaces enabled making use of the caretakers' experiences and reflections on the system's difficulties and challenges (and even the misuses).



Figure 8: Example of co-creating possible scenarios with persona descriptions as input.

Furthermore, these stakeholder workshops were also considered learning spaces. The intervention was directed not only at the employees, those who engaged in the interventions as caretakers or daytime users, but also at the employers, the service provider, and the vehicle manufacturer. These stakeholders appreciated the learning spaces provided by academia. However, even though all the stakeholders in this research project agreed on the overall need for sustainability transitions, they needed to complete more work to decide what should be done next to test any updated system.

CASE 3

The third case is a research project with a Swedish corporation aiming to upgrade its corporate mobility service system. This included commuter buses, shuttle buses, and taxis supporting transportation at the work site (see Figure 9). For more details regarding the research project Sustainable Mobility Services Södertälje, see Hesselgren et al. (2020).

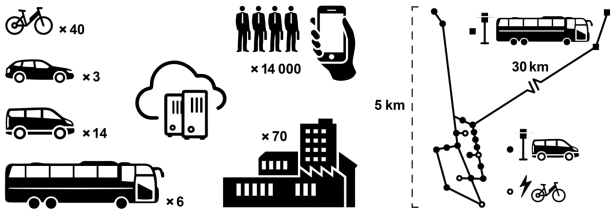


Figure 9: Overview of the Corporate Mobility Service System.

This project aimed to redesign the existing service system by introducing a digital application showing real-time data of all vehicle positions, thereby connecting the system. Moreover, the company's purpose was to support the shift towards more sustainable transport practices amongst the employees, thereby supporting the company's sustainability vision. For research purposes, there was also an aim to disrupt practices to evoke reflections among employees on their commuting and travel practices. Therefore, to reconfigure the service system, it was decided to include a new vehicle, shared electric bikes, to complement the existing vehicles.

To redesign the service system, employees were invited to share their experiences of the existing one. This was done through short contextual on-site and longer semi-structured interviews in three iterative development loops. Travel diaries were used as sensitising material in one of the loops to prepare the interviewees by encouraging them to reflect on their commuting and travel practices, thereby developing insights. Future scenarios (see Figure 10) were introduced as trigger material in another of the loops to spark discussions and provoke imaginations of what the service system could potentially be.



Figure 10: Example of scenario as trigger material.

Some employees developed new mobility practices when the reconfigured system was launched with the added shared electric bikes (see Figure 11). Commuting and on-site transport practices changed to some extent,

and proto-practices developed. Even those who had never used an electric bike found it easy. It was clear that there were no difficulties in acquiring the related practices such as booking, starting, running, and returning the bikes. Moreover, electric bike users also discovered many additional benefits.

Some employees said that they appreciated the flexibility of using the bikes instead of having to depend on the timetables of the buses. Furthermore, some employees considered themselves to become more time-efficient regarding their work commitments. Freedom and time efficiency are often connected to car driving, but in this case, they were also related to bike usage. Others paid more attention to getting daylight and fresh air during their workdays, which could help them relax and feel more creative. Such values are typically connected to cycling and did not come as any surprise.

Additionally, some employees expressed that they valued the excitement of riding electric bikes and enjoyed the fun. No other transport mode in the corporate mobility service system was described as fun and exciting. The other vehicle modes were considered functional. Having fun is not a typical corporate value but could be considered essential to promote as an entry point for potential users to engage in this mobility practice.



Figure 11: The shared electric bikes.

To further understand how to develop the service system and gain insights into the employees' mobility practices, attention was paid to how the electric bikes were used outside the intentions. In stakeholder workshops that were designed as learning spaces, there was a particular focus on misuses as potentials for development. Attention was paid to how the users reconfigured the system, and the alternative uses that appeared as the new mobility practices developed.

Initially, the intention was that electric bikes should be used between four dedicated office buildings during work hours. Thereby, they should facilitate transport between meetings in these buildings. However, in the interviews, several employees expressed that they preferred to use electric bikes in conjunction with arriving or departing with commuter trains. The company had not advertised this as possible since Swedish tax regulations prevented the internal corporate mobility system from connecting to public transport. If these two systems had been connected, all employees would have been subject to benefit taxation. However, this is precisely what the employees did – they connected the two systems since they needed to align their work trips on-site with their commuting trips since that was a practical way of organising everyday life. Therefore, as discussed in stakeholder workshops, different strategies could be used to overcome such glitches. Future developments could be suggested by focusing on glitches as a design material. To up-scale the system could mean that more electric bikes should be brought in to increase availability, more docking stations could be added, and longer operating hours could be introduced to make the system even more accessible. However, it could also mean changing the system rules to allow employees to connect their commuting to internal work trips.

DISCUSSION

To design for sustainability transitions means that doing the design can be considered more important than the design outcome. As described here, the RE-DO Design process used in the interventions focuses on enabling learning spaces, supporting collaborative sensemaking, and promoting ongoing transformations. These perspectives on designing are discussed to answer the question: *What re-formulations of design as a practice are needed when designing conditions for sustainable mobility practices and supporting transformative learning?* In the interventions, designing was used to detect and amplify what might be taken for granted. Moreover, through careful analysis, designing was used to reformulate more sustainable options for everyday living.

INTERVENTIONS AS LEARNING SPACES

In the described mobility interventions, there are several aspects of how learning took place. First, for the participants who changed their practices, learning involved developing insights and engaging them in communities of practice (Lave, 1991; Lave & Wenger, 1991). Here, the participants learned from their own mobility practices and from others who engaged in these

practices and shared their experiences. Such learning is always situated. Moreover, the participants were also encouraged to reflect on their mobility practices. Thereby, both bodies and minds were engaged, and together with reflections on the tacit knowledge that developed while engaged, this assisted deep learning. Supporting practical knowledge to develop is essential when striving to engage people in sustainability transitions (Sahakian & Wilhite, 2014). In the interventions, this was done by reconfiguring practices through the introduction of new material elements (new vehicles) and thereby provoking a "crisis of routines" (Reckwitz, 2002) to engage the participants in proto-practices. As such, the materials acted as learning devices, and when introduced in interventions, they formed parts of practices. The practical knowledge the participants developed was captured, for example, in the travel diaries, interviews and workshops.

Second, when the participants were engaged in the interventions and learned from their experiences, this can also be considered double-loop learning. However, to support such learning as a designer, it becomes essential to use suitable design strategies (Hesselgren, 2019). This type of design for sustainability is focused on processes rather than outcomes. By designing learning spaces in the interventions, with process attention, the participants, such as the family members enrolling in car-free living and the employees trying new mobility practices, gained practical knowledge from their proto-practices. Furthermore, other participants and stakeholders in the interventions, such as the decision-makers, learned from the participants' practices through the researchers' capturing and exploring activities. The double-loop learning was supported by sensitising materials, such as reflective questions in the travel diaries and preparatory assignments before interviews and workshops. These design research materials were designed as learning devices that were introduced at different stages in the interventions.

Third, when bridging citizens' learnings to decision-makers, third-loop learning can occur when there are reflections on the actual learning processes while engaging in the proto-practices. Learning devices can be materials, but future scenarios that trigger discussions can also be considered learning devices. Spaces where participatory learning occurs sometimes have conflicts, whereby a designer's role can be to shed light on such tensions. In these learning spaces, future scenarios can be presented to decision-makers. These scenarios can act as "learning machines" (Berkhout et al., 2002). Introducing such scenarios can have the deliberate purpose of engaging stakeholders in unlearning and re-learning (Robinson, 2003). In the interventions, different examples of future scenarios were introduced as learning devices, for example, the supporting concepts for car-free living. These were purposely developed to illuminate and challenge given categories, for example, congestion charges for cars and incentives for cycling. When these were discussed with local, regional, and national authorities in the bridging stage, they generated discussions on possible solutions and how the stakeholders could work with more project initiatives to understand citizens' perspectives better.

INTERVENTIONS AS CO-SENSEMAKING

As sustainable mobility practices were formed, the participants connected the materials with new skills in the mobility interventions. They also, in some cases, created new meanings around, for example, parenthood, citizenship and employee agreements. To understand the values important to the participants, it was necessary to gain insights into how new meanings were formed and how the participants made sense of their activities.

In sensemaking and sensegiving processes, understanding the means needed to reach an outcome or why something occurred can be essential communication when motivating decisions (Hill & Levenhagen, 1995; Weick & Sutcliffe, 2005; Gioia & Chittipeddi, 1991). When actors individually or collectively participate in sensemaking and sensegiving processes (Stigliani & Ravasi, 2012), they attach meaning to innovations (Verganti & Öberg, 2013). Such innovation processes can be considered as results of interpreting and envisioning, also radically new meanings (ibid.).

The design processes focused on interpreting and envisioning new meanings in the three mobility interventions. In the exploring and bridging stages, the interpreted values of the mobility service systems were pinpointed. These processes took place with the participants of the interventions while exploring and with stakeholders involved in the research projects in the bridging stages. Thereby, sensemaking was a collaborative process, and the interventions were as such spaces for co-sensemaking to take place. For example, meanings associated with electric vehicles, discovered through performing proto-practices with these, were discussed in co-design workshops in all three mobility interventions in the bridging stages. Here, the meanings the participants expressed they had discovered were used as input for further discussions on possible upscaling and re-orientation of existing practices. For example, the meaning of being an efficient employee or sustainable citizen was explored.

In the bridging stage, alternative actions for changing conditions for sustainable mobility practices were explored. Representing the participants' experiences through the concepts and future scenarios worked well to provoke discussions, and as such, they worked well as re-orienting devices and learning probes. However,

some design concepts were, in some contexts, too provocative for the workshop participants to make sense of current urban planning or existing corporate culture. Moreover, since all three interventions were time-limited research projects, even though the learning probes worked well to envision potential future possibilities, this did not always necessarily lead to decisions on future actions.

INTERVENTIONS AS TRANSFORMATIVE

Design practices, including service design, can bring tools, methods, and structure to innovation processes (Korper et al., 2020). One aspect of design practice is that it can assist participation and co-creation. Furthermore, designing can bring possibilities for making sense together. The design practice used in the described interventions focused on processes of experimentation and learning from these experiments to support ongoing transformation. To some extent, the purpose of the interventions was thereby no longer the interventions in themselves but instead to initiate transformative learning. Thereby, designing interventions are collaborative processes, where design practice is used to build knowledge and initiate actions. Sensing and envisioning collaboratively and co-shaping sustainable mobility solutions are essential to designing conditions for sustainable mobility practices (Hesselgren, 2019). Furthermore, such design for sustainability approach is at the socio-technical system level where design tools and methods support integrative and collaborative competencies to develop.

In the interventions, the design sensing and giving processes were used to detect and amplify what might be taken for granted. Through analysis, it was possible to reformulate everyday living and suggest more sustainable possibilities. Moreover, with thoughtful considerations, the design suggestions built as the bridging activities proposed alternative configurations of humans and technologies. With practices as design material, this was a way to get to everyday life and to design both structure and lifestyle elements whereby conditions for practices could be shaped for continuously ongoing everyday life.

In designing the trigger material and future scenarios introduced in the interventions, there was a balance between provocative and affirmative design (Hesselgren et al., 2017). On the one hand, it was necessary to stir up emotions and thoughts on what could be possible, and at the same time, connections to recognisable everyday situations were needed. It was with such balancing design strategies that transformative learning could be supported. Thereby, the interventions could be part of ongoing change processes.

CONCLUSIONS

To design for sustainability, design as a practice needs an altered position. As a complement to creating sleek products or seamless services, attention to designing learning spaces is necessary, and designing such learning spaces requires careful crafting. Moreover, the design needs to focus on ongoing processes, where the design is made and continuously re-made. RE-DO Design Doings are proposed as a design practice for designing conditions for sustainable mobility practices and supporting transformative learning.

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